

REQUIRMENT SPECIFIC / PRELIMINARIES

3.1 Existing System

Traditional radiation monitoring setups mainly depend on handheld radiation detectors or fixed monitoring stations, to register radiation levels. Even if these methods give accurate radiation readings, they still don't really provide a real-time geospatial picture, or any steady environmental tracking. In practice, the radiation information is often gathered by hand, and then processed on its own schedule. Because of that, it becomes hard to pinpoint the exact spot of radioactive sources during on-site field work. Also, the typical approach does not neatly bring together radiation sensors, GPS modules, environmental sensors, and the visualization platform, so the whole system ends up less effective when you need broad-area radiation surveillance or fast emergency response.

3.2 Problem Statement

The biggest problem in the radiation monitoring systems we already have is basically that they cannot show the radiation distribution in real time while still, accurately pin pointing where the source is. The standard 2D radiation heatmaps tend to only give radiation intensity across a surface, but they do not really add depth, or the spatial context you would want in the field. Also there is no real integrated **GPS telemetry**, no environmental sensing, and no live visualization, so during nuclear checks or emergency events, it becomes harder to make the right decision quickly. So what is needed is a smarter setup that can do real-time radiation charting, keep an eye on the environment, and later also support localization of the radioactive source.

3.3 Proposed Solution

The proposed solution looks like **RADMAP**, a kind GPS-enabled radiation mapping prototype running on a **Raspberry Pi-5**, it's meant to be practical but also a bit experimental. In the setup, the system pulls together a Gamma Spectrometer, a **GlobalSat BU-353N GPS receiver**, a **DHT22 environmental sensor**, and a camera module, so it can grab radiation counts plus GPS coordinates, along with temperature and humidity and even live video at the same time, kind of altogether.

After that, the captured information goes into a Python-based backend where it's processed into interactive **2D radiation heatmaps**, and also **3D Digital Twin**, for clearer visualization, and for helping with radioactive source localization later on. For the live feed, a color-grading algorithm, inspired by night-vision techniques, is used to show different radiation intensity levels right on the camera view, so it becomes easier to spot radiation hotspots without too much guessing. The architecture is modular too, meaning the design can expand later with AI-based '**SLAM**' **source localization**, **LiDAR integration**, autonomous robotic navigation, and cloud-based radiation monitoring too, if needed.

4.4 Requirement Specifications

Component	Specification
Raspberry Pi 5	8 GB RAM
GlobalSat BU-353N	GPS Receiver
CNSPEC MCA	Gamma Spectrometer
DHT22	Temperature & Humidity Sensor
Camera Module	Live Monitoring

Table 1: Hardware Requirements

Software	Purpose
Raspberry Pi OS	Operating System
Python 3.x	Backend Development
Streamlit	Dashboard
OpenCV	Camera Processing
Plotly	3D Visualization
Folium	2D Heatmap
Pandas	Data Processing
NumPy	Numerical Computing
GitHub	Version Control (VC)

Table 2: Software Requirements